

## Online Resource 1

Supplementary tables, figure and questionnaire for: Internal Control Architectures and Accountability Dimensions in Emerging Economy Bureaucracies

### S1 Sample and measurement

**Table S1** Demographic characteristics of the respondents (N = 187)

| Variable                | Category          | Frequency | Percentage (%) | Cumulative (%) |
|-------------------------|-------------------|-----------|----------------|----------------|
| Gender                  | Male              | 126       | 67.4           | 67.4           |
|                         | Female            | 61        | 32.6           | 100.0          |
| Age                     | 20–25 years       | 19        | 10.2           | 10.2           |
|                         | 26–30 years       | 53        | 28.3           | 38.5           |
|                         | 31–35 years       | 50        | 26.7           | 65.2           |
|                         | Over 35 years     | 65        | 34.8           | 100.0          |
| Level of Education      | Diploma           | 1         | 0.5            | 0.5            |
|                         | Bachelor's Degree | 18        | 9.6            | 10.2           |
|                         | Master's Degree   | 92        | 49.2           | 59.4           |
|                         | Ph.D.             | 76        | 40.6           | 100.0          |
| Professional Experience | Less than 5 years | 41        | 21.9           | 21.9           |
|                         | 6–10 years        | 72        | 38.5           | 60.4           |
|                         | 11–15 years       | 43        | 23.0           | 83.4           |
|                         | Over 15 years     | 31        | 16.6           | 100.0          |

Note. Source: survey data.

**Table S2** Item statistics and CFA loadings (N = 187)

| Item | Construct | Mean  | SD    | Skewness | Kurtosis | $\lambda$ | b     | SE    | t      | $\lambda^2$ |
|------|-----------|-------|-------|----------|----------|-----------|-------|-------|--------|-------------|
| Q1   | CA        | 3.481 | 0.986 | -0.543   | 0.076    | 0.754     | 1.000 | –     | –      | 0.569       |
| Q2   | CA        | 3.401 | 1.050 | -0.243   | -0.546   | 0.755     | 1.066 | 0.100 | 10.692 | 0.570       |
| Q3   | CA        | 3.460 | 1.054 | -0.479   | -0.303   | 0.820     | 1.162 | 0.099 | 11.753 | 0.672       |
| Q4   | CA        | 3.455 | 1.074 | -0.355   | -0.514   | 0.790     | 1.141 | 0.101 | 11.266 | 0.624       |
| Q5   | CA        | 3.417 | 0.999 | -0.326   | -0.244   | 0.727     | 0.977 | 0.095 | 10.245 | 0.529       |
| Q6   | CA        | 3.647 | 0.986 | -0.468   | -0.098   | 0.797     | 1.057 | 0.093 | 11.374 | 0.635       |
| Q7   | CA        | 3.508 | 1.007 | -0.549   | -0.019   | 0.722     | 0.979 | 0.096 | 10.169 | 0.522       |
| Q8   | CA        | 3.401 | 0.942 | -0.335   | -0.213   | 0.796     | 1.008 | 0.089 | 11.357 | 0.633       |
| Q9   | CA        | 3.503 | 0.991 | -0.309   | -0.333   | 0.807     | 1.076 | 0.093 | 11.543 | 0.651       |
| Q10  | FM        | 3.701 | 1.019 | -0.726   | 0.198    | 0.677     | 1.000 | –     | –      | 0.459       |
| Q11  | FM        | 3.674 | 1.008 | -0.453   | -0.349   | 0.748     | 1.093 | 0.130 | 8.419  | 0.559       |
| Q12  | FM        | 3.567 | 0.950 | -0.289   | -0.348   | 0.718     | 0.989 | 0.121 | 8.174  | 0.516       |
| Q13  | FM        | 3.508 | 0.964 | -0.406   | -0.160   | 0.678     | 0.947 | 0.121 | 7.810  | 0.460       |
| Z1   | ORG       | 3.882 | 0.960 | -0.609   | -0.192   | 0.816     | 1.000 | –     | –      | 0.667       |
| Z2   | ORG       | 3.642 | 1.055 | -0.462   | -0.263   | 0.856     | 1.153 | 0.081 | 14.153 | 0.733       |
| Z3   | ORG       | 3.567 | 1.102 | -0.415   | -0.418   | 0.834     | 1.173 | 0.086 | 13.601 | 0.696       |
| Z4   | ORG       | 3.471 | 1.038 | -0.430   | -0.064   | 0.830     | 1.099 | 0.081 | 13.493 | 0.688       |
| Z5   | ORG       | 3.695 | 1.036 | -0.563   | -0.274   | 0.826     | 1.092 | 0.081 | 13.402 | 0.682       |
| Z6   | ORG       | 3.535 | 1.137 | -0.430   | -0.635   | 0.873     | 1.267 | 0.087 | 14.587 | 0.763       |
| Z7   | ORG       | 3.877 | 0.995 | -0.544   | -0.457   | 0.807     | 1.024 | 0.079 | 12.945 | 0.651       |
| Z8   | LEG       | 3.572 | 1.062 | -0.246   | -0.750   | 0.684     | 1.000 | –     | –      | 0.468       |
| Z9   | LEG       | 3.358 | 1.095 | -0.353   | -0.536   | 0.871     | 1.313 | 0.123 | 10.670 | 0.759       |

| Item | Construct | Mean  | SD    | Skewness | Kurtosis | $\lambda$ | b     | SE    | t      | $\lambda^2$ |
|------|-----------|-------|-------|----------|----------|-----------|-------|-------|--------|-------------|
| Z10  | LEG       | 3.439 | 1.112 | -0.223   | -0.796   | 0.768     | 1.176 | 0.123 | 9.572  | 0.590       |
| Z11  | LEG       | 3.465 | 1.028 | -0.101   | -0.669   | 0.836     | 1.184 | 0.115 | 10.316 | 0.700       |
| Z12  | LEG       | 3.513 | 0.991 | -0.506   | -0.032   | 0.813     | 1.110 | 0.110 | 10.070 | 0.662       |
| Z13  | LEG       | 3.508 | 1.007 | -0.453   | -0.151   | 0.685     | 0.950 | 0.110 | 8.627  | 0.470       |
| Z14  | PRO       | 3.749 | 1.009 | -0.431   | -0.608   | 0.733     | 1.000 | –     | –      | 0.538       |
| Z15  | PRO       | 3.529 | 0.980 | -0.222   | -0.391   | 0.800     | 1.060 | 0.097 | 10.966 | 0.641       |
| Z16  | PRO       | 3.829 | 0.929 | -0.587   | 0.129    | 0.774     | 0.972 | 0.092 | 10.584 | 0.600       |
| Z17  | PRO       | 3.701 | 1.003 | -0.467   | -0.168   | 0.836     | 1.134 | 0.099 | 11.493 | 0.700       |
| Z18  | PRO       | 3.374 | 1.135 | -0.219   | -0.827   | 0.874     | 1.342 | 0.111 | 12.040 | 0.764       |
| Z19  | PRO       | 3.583 | 1.046 | -0.379   | -0.410   | 0.721     | 1.020 | 0.104 | 9.812  | 0.520       |
| Z20  | PRO       | 3.604 | 1.049 | -0.534   | -0.346   | 0.797     | 1.130 | 0.104 | 10.910 | 0.635       |
| Z21  | POL       | 3.439 | 1.078 | -0.139   | -0.810   | 0.803     | 1.000 | –     | –      | 0.645       |
| Z22  | POL       | 3.422 | 1.020 | -0.217   | -0.582   | 0.789     | 0.930 | 0.081 | 11.525 | 0.623       |
| Z23  | POL       | 3.460 | 1.038 | -0.169   | -0.729   | 0.792     | 0.950 | 0.082 | 11.574 | 0.627       |
| Z24  | POL       | 3.273 | 1.070 | -0.298   | -0.358   | 0.834     | 1.032 | 0.084 | 12.306 | 0.696       |

Note.  $\lambda$  = standardized loading; b = unstandardized loading; SE = standard error; t = critical ratio;  $\lambda^2$  = item reliability. The first loading of each construct (b = 1.000) was fixed. All free loadings are significant at  $p < 0.001$ . Skewness and kurtosis are bias-corrected.

## S2 Distributional properties

**Table S3** Univariate normality tests of the constructs

| Construct            | K–S D | K–S p | Lilliefors p | S–W W | S–W p   | z skewness | z kurtosis |
|----------------------|-------|-------|--------------|-------|---------|------------|------------|
| Control Activities   | 0.072 | 0.268 | 0.023        | 0.977 | 0.004   | -2.757     | 0.484      |
| Financial Management | 0.077 | 0.203 | 0.011        | 0.973 | 0.001   | -2.274     | -0.025     |
| Organizational       | 0.081 | 0.162 | 0.006        | 0.962 | < 0.001 | -2.897     | -0.793     |
| Legal                | 0.075 | 0.236 | 0.018        | 0.974 | 0.001   | -1.097     | -2.298     |
| Professional         | 0.069 | 0.325 | 0.040        | 0.975 | 0.002   | -2.149     | -0.891     |
| Political            | 0.095 | 0.063 | 0.001        | 0.977 | 0.004   | -0.612     | -1.896     |

Note. K–S = Kolmogorov–Smirnov; S–W = Shapiro–Wilk. Shapiro–Wilk rejected normality for all 37 items ( $p < 0.001$ ).

**Table S4** Multivariate normality (Mardia’s test)

| Coefficient                                                | Statistic | Expected value / df | Test value | p     |
|------------------------------------------------------------|-----------|---------------------|------------|-------|
| Multivariate skewness ( $\chi^2$ )                         | 301.033   | 9,139               | 9,382.183  | 0.037 |
| Multivariate skewness ( $\chi^2$ , small-sample corrected) | 301.033   | 9,139               | 9,540.712  | 0.002 |
| Multivariate kurtosis (z)                                  | 1,444.832 | 1,443               | 0.233      | 0.816 |

Note. The small-sample corrected skewness statistic uses the correction of Mardia (1970).

## S3 Validity

**Table S5** Fornell–Larcker criterion (diagonal = square root of AVE; below = latent correlations)

|     | CA           | FM           | ORG          | LEG          | PRO          | POL          |
|-----|--------------|--------------|--------------|--------------|--------------|--------------|
| CA  | <b>0.775</b> |              |              |              |              |              |
| FM  | 0.500        | <b>0.706</b> |              |              |              |              |
| ORG | 0.540        | 0.569        | <b>0.835</b> |              |              |              |
| LEG | 0.487        | 0.376        | 0.397        | <b>0.780</b> |              |              |
| PRO | 0.575        | 0.471        | 0.321        | 0.317        | <b>0.792</b> |              |
| POL | 0.515        | 0.597        | 0.388        | 0.411        | 0.426        | <b>0.805</b> |

Note. Square roots of AVE are in bold.

**Table S6** HTMT (below the diagonal) and HTMT2 (above the diagonal)

|     | CA    | FM    | ORG   | LEG   | PRO   | POL   |
|-----|-------|-------|-------|-------|-------|-------|
| CA  | –     | 0.508 | 0.541 | 0.501 | 0.567 | 0.512 |
| FM  | 0.513 | –     | 0.561 | 0.370 | 0.446 | 0.594 |
| ORG | 0.544 | 0.569 | –     | 0.399 | 0.289 | 0.379 |
| LEG | 0.506 | 0.381 | 0.405 | –     | 0.304 | 0.391 |
| PRO | 0.573 | 0.467 | 0.308 | 0.320 | –     | 0.411 |
| POL | 0.516 | 0.596 | 0.391 | 0.402 | 0.425 | –     |

Note. Bootstrap 95% upper limits of HTMT (2,000 resamples) range from 0.454 to 0.707, all below 0.85.

**Table S7** Common method bias diagnostics

| Procedure                                             | Result                                                               | Criterion / comparison                                        | Assessment                                                    |
|-------------------------------------------------------|----------------------------------------------------------------------|---------------------------------------------------------------|---------------------------------------------------------------|
| Harman's single-factor test                           | First factor explains 36.02% of variance                             | < 50%                                                         | Passed                                                        |
| One-factor CFA                                        | $\chi^2(629) = 2,803.267$ ; CFI = 0.522; RMSEA = 0.136; SRMR = 0.128 | Six-factor model: $\Delta\chi^2(15) = 2,077.55$ , $p < 0.001$ | Single factor rejected                                        |
| CLF model, free method loadings                       | $\chi^2(577) = 650.143$ ; CFI = 0.984; RMSEA = 0.026; SRMR = 0.044   | Six-factor model: $\Delta\chi^2(37) = 75.58$                  | Degenerate solution; 28.42% shared variance not interpretable |
| CLF model, equal method loadings                      | $\chi^2(613) = 723.687$ ; CFI = 0.976; RMSEA = 0.031; SRMR = 0.051   | Six-factor model: $\Delta\chi^2(1) = 2.03$ , $p = 0.154$      | Fit not improved                                              |
| Variance shared by the method factor (equal loadings) | 12.48%                                                               | < 25%                                                         | Below benchmark                                               |

Note. The free-loading CLF solution was degenerate (variances of one or more constructs near zero), so its shared-variance estimate is not interpretable.

## S4 Structural model

**Table S8** Bootstrap confidence intervals (B = 2,000) and effect sizes of the structural paths

| Hypothesis | Path                 | $\beta$ | Boot. SE | 95% CI of $\beta$ | $b$   | 95% CI of $b$  | $f^2$ | Effect size |
|------------|----------------------|---------|----------|-------------------|-------|----------------|-------|-------------|
| H1         | CA $\rightarrow$ ORG | 0.342   | 0.078    | [0.179, 0.491]    | 0.360 | [0.182, 0.557] | 0.147 | Small       |
| H2         | CA $\rightarrow$ LEG | 0.390   | 0.087    | [0.214, 0.557]    | 0.382 | [0.200, 0.593] | 0.157 | Medium      |
| H3         | CA $\rightarrow$ PRO | 0.452   | 0.082    | [0.280, 0.601]    | 0.449 | [0.275, 0.628] | 0.244 | Medium      |
| H4         | CA $\rightarrow$ POL | 0.290   | 0.077    | [0.137, 0.437]    | 0.338 | [0.159, 0.529] | 0.110 | Small       |
| H5         | FM $\rightarrow$ ORG | 0.394   | 0.080    | [0.232, 0.544]    | 0.445 | [0.262, 0.650] | 0.196 | Medium      |
| H6         | FM $\rightarrow$ LEG | 0.207   | 0.093    | [0.024, 0.385]    | 0.217 | [0.026, 0.423] | 0.044 | Small       |
| H7         | FM $\rightarrow$ PRO | 0.243   | 0.082    | [0.085, 0.399]    | 0.259 | [0.081, 0.460] | 0.071 | Small       |
| H8         | FM $\rightarrow$ POL | 0.457   | 0.084    | [0.294, 0.607]    | 0.570 | [0.345, 0.863] | 0.273 | Medium      |

Note.  $f^2 = \beta^2(1 - r^2)/(1 - R^2)$ ,  $r = 0.50$ .

**Table S9** Bootstrap tests of the difference between the effects of control activities and financial management

| Outcome                    | $\beta$ (CA) | $\beta$ (FM) | $\Delta\beta$ (CA – FM) | Boot. SE | 95% CI of $\Delta\beta$ | $p$ (bootstrap) | Significant difference |
|----------------------------|--------------|--------------|-------------------------|----------|-------------------------|-----------------|------------------------|
| Organizational (H1 vs. H5) | 0.342        | 0.394        | -0.052                  | 0.146    | [-0.342, 0.232]         | 0.697           | No                     |
| Legal (H2 vs. H6)          | 0.390        | 0.207        | 0.183                   | 0.167    | [-0.150, 0.498]         | 0.253           | No                     |
| Professional (H3 vs. H7)   | 0.452        | 0.243        | 0.209                   | 0.151    | [-0.091, 0.492]         | 0.166           | No                     |
| Political (H4 vs. H8)      | 0.290        | 0.457        | -0.167                  | 0.148    | [-0.450, 0.111]         | 0.268           | No                     |

Note.  $\Delta\beta$  computed in each of 2,000 resamples; two-sided bootstrap  $p$ -values.

**Table S10** Explained variance ( $R^2$ ) of the endogenous constructs

| Endogenous construct | $R^2$ (latent, SEM) | $R^2$ (composite, OLS) | Adjusted $R^2$ | $F(2, 184)$ | $p$     |
|----------------------|---------------------|------------------------|----------------|-------------|---------|
| Organizational (ORG) | 0.407               | 0.348                  | 0.341          | 49.184      | < 0.001 |
| Legal (LEG)          | 0.275               | 0.233                  | 0.225          | 28.019      | < 0.001 |
| Professional (PRO)   | 0.374               | 0.316                  | 0.309          | 42.527      | < 0.001 |
| Political (POL)      | 0.426               | 0.326                  | 0.318          | 44.416      | < 0.001 |

Note. Composite R<sup>2</sup> from ordinary least squares on construct scores.

**Table S11** Comparison of competing structural models

| Model                                                  | $\chi^2$ | df  | CFI   | RMSE A | SRMR  | AIC     | BIC     | $\Delta\chi^2$ vs. proposed | $\Delta df$ | p ( $\Delta\chi^2$ ) |
|--------------------------------------------------------|----------|-----|-------|--------|-------|---------|---------|-----------------------------|-------------|----------------------|
| Proposed model (all eight paths)                       | 733.18   | 620 | 0.975 | 0.031  | 0.052 | 899.2   | 1,167.4 | –                           | –           | –                    |
| Control activities only (FM paths fixed to zero)       | 789.91   | 624 | 0.964 | 0.038  | 0.072 | 947.9   | 1,203.2 | 56.73                       | 4           | < 0.001              |
| Financial management only (CA paths fixed to zero)     | 791.23   | 624 | 0.963 | 0.038  | 0.075 | 949.2   | 1,204.5 | 58.05                       | 4           | < 0.001              |
| Null model (no paths; endogenous factors uncorrelated) | 1,010.70 | 628 | 0.916 | 0.057  | 0.256 | 1,160.7 | 1,403.0 | 277.52                      | 8           | < 0.001              |
| Saturated model (freely correlated factors = CFA)      | 725.72   | 614 | 0.975 | 0.031  | 0.048 | 903.7   | 1,191.3 | 7.46                        | 6           | 0.280                |

Note. AIC and BIC are computed from  $\chi^2$ ; only differences are meaningful.

**Table S12** Sensitivity of the standardized path coefficients ( $\beta$ ) across specifications

| Path        | Base model, ML | DWLS (Pearson)     | Ordinal (polychoric) <sup>a</sup> | Outliers removed (N = 177) | Four correlated errors | Correlated disturbances | Method factor controlled | Demographic controls |
|-------------|----------------|--------------------|-----------------------------------|----------------------------|------------------------|-------------------------|--------------------------|----------------------|
| H1 CA → ORG | 0.342***       | 0.339***           | 0.318                             | 0.314***                   | 0.342***               | 0.341***                | 0.311***                 | 0.346***             |
| H2 CA → LEG | 0.390***       | 0.398***           | 0.379                             | 0.397***                   | 0.392***               | 0.399***                | 0.308**                  | 0.377***             |
| H3 CA → PRO | 0.452***       | 0.452***           | 0.464                             | 0.465***                   | 0.449***               | 0.452***                | 0.395***                 | 0.461***             |
| H4 CA → POL | 0.290***       | 0.303***           | 0.296                             | 0.293**                    | 0.289***               | 0.289***                | 0.252**                  | 0.280***             |
| H5 FM → ORG | 0.394***       | 0.385***           | 0.408                             | 0.424***                   | 0.395***               | 0.399***                | 0.347***                 | 0.397***             |
| H6 FM → LEG | 0.207*         | 0.248**            | 0.257                             | 0.221*                     | 0.204*                 | 0.176 ns                | 0.091 ns                 | 0.213*               |
| H7 FM → PRO | 0.243**        | 0.217*             | 0.222                             | 0.233*                     | 0.245**                | 0.245**                 | 0.178 ns                 | 0.236**              |
| H8 FM → POL | 0.457***       | 0.464***           | 0.462                             | 0.430***                   | 0.459***               | 0.452***                | 0.395***                 | 0.456***             |
| CFI         | 0.975          | 1.000 <sup>b</sup> | –                                 | 0.973                      | 0.978                  | 0.975                   | 0.976                    | 0.973                |
| RMSEA       | 0.031          | 0.000 <sup>b</sup> | –                                 | 0.033                      | 0.029                  | 0.031                   | 0.031                    | 0.030                |
| SRMR        | 0.052          | 0.047              | 0.053                             | 0.051                      | 0.052                  | 0.048                   | 0.055                    | 0.050                |

Note. \*\*\* p < 0.001, \*\* p < 0.01, \* p < 0.05, ns = not significant. a Point estimates only. b DWLS indices truncated at their bounds.

## S5 Figure

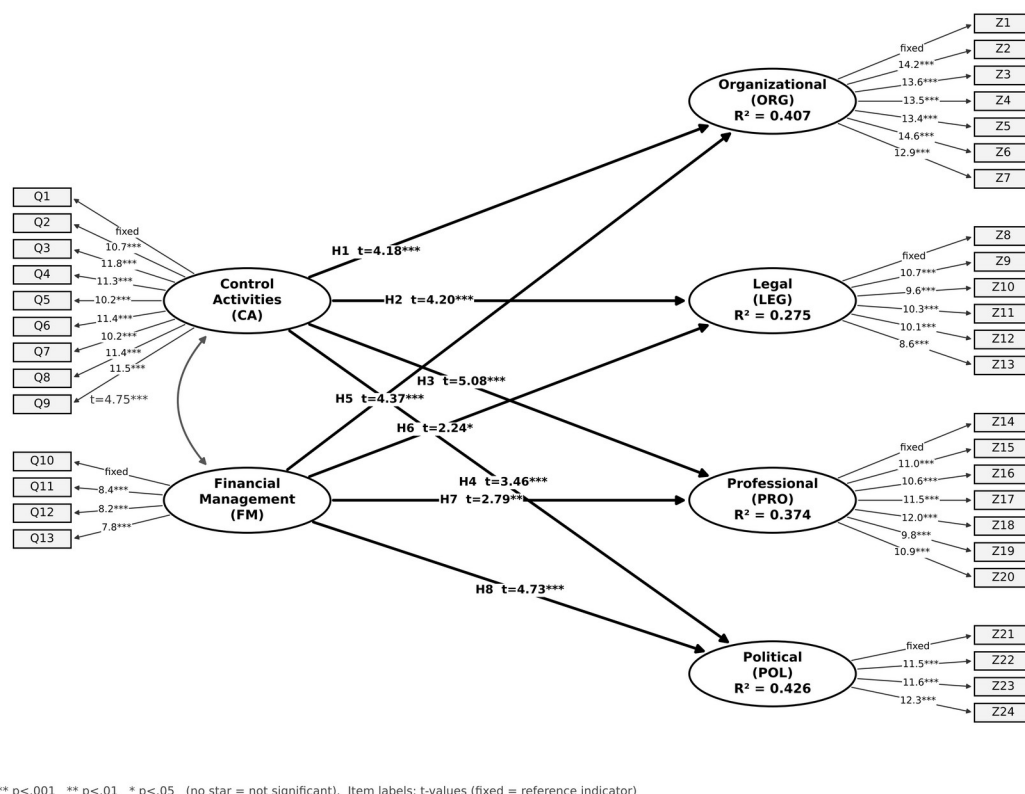


Fig. S1 Estimated structural model with t-values (fixed = reference indicator).

## S6 Questionnaire

All items were rated from 1 (very little) to 5 (very much). Part A recorded gender (male, female), age (20–25, 26–30, 31–35, over 35 years), education (diploma, bachelor's degree, master's degree, Ph.D.) and professional experience (less than 5, 6–10, 11–15, over 15 years). Parts B and C were introduced as follows: “Please indicate the degree to which these financial control mechanisms are present or enforced” and “Below are questions evaluating your current perspective on the diverse dimensions of accountability within the Tax Affairs Administration.”

Table S13 Questionnaire items

| Item | Construct          | Question                                                                                                                |
|------|--------------------|-------------------------------------------------------------------------------------------------------------------------|
| Q1   | Control activities | To what extent is there a clear segregation of roles and responsibilities in the organization?                          |
| Q2   | Control activities | To what extent is each individual's duty independently reviewed by another staff member?                                |
| Q3   | Control activities | To what extent is there adequate supervision provided by senior management?                                             |
| Q4   | Control activities | To what extent are prompt corrective actions taken to address identified weaknesses in internal controls?               |
| Q5   | Control activities | To what extent does the organization employ adequately trained personnel for implementing financial management systems? |
| Q6   | Control activities | To what extent is a detailed and standardized chart of accounts available and utilized?                                 |
| Q7   | Control activities | To what extent is every organizational expenditure strictly confined within the approved budget?                        |

| Item | Construct                     | Question                                                                                                                            |
|------|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| Q8   | Control activities            | To what extent are budget discrepancy reports, accompanied by detailed explanations, routinely prepared?                            |
| Q9   | Control activities            | To what extent is there an appropriate and reliable security system for safeguarding organizational assets?                         |
| Q10  | Financial management          | To what extent are sufficient financial resources available to meet organizational obligations?                                     |
| Q11  | Financial management          | To what extent are salaries and wages structured appropriately and based on professional merit?                                     |
| Q12  | Financial management          | To what extent have unnecessary or wasteful expenditures been successfully reduced within the organization?                         |
| Q13  | Financial management          | To what extent have outstanding organizational invoices and liabilities been systematically settled?                                |
| Z1   | Organizational accountability | To what extent are citizen visits and administrative procedures optimized for efficient service delivery?                           |
| Z2   | Organizational accountability | To what extent do managers and staff consistently fulfill their professional commitments and promises?                              |
| Z3   | Organizational accountability | To what extent do staff members exhibit respectful and professional behavior toward citizens?                                       |
| Z4   | Organizational accountability | To what extent are nepotism and administrative bribery effectively prevented in this department?                                    |
| Z5   | Organizational accountability | To what extent are ethical and human values adhered to in the daily behavior of the staff?                                          |
| Z6   | Organizational accountability | To what extent are managers and administrative staff accessible to the public?                                                      |
| Z7   | Organizational accountability | To what extent is excessive bureaucracy (red tape) and unnecessary paperwork minimized?                                             |
| Z8   | Legal accountability          | To what extent are statutory laws and organizational regulations correctly and strictly implemented?                                |
| Z9   | Legal accountability          | To what extent is prompt and accurate information regarding new laws and regulations provided to the public?                        |
| Z10  | Legal accountability          | To what extent are legal loopholes and citizens' regulatory grievances actively addressed?                                          |
| Z11  | Legal accountability          | To what extent is the department's annual budget transparently presented or made accessible to the public?                          |
| Z12  | Legal accountability          | To what extent is the execution of overly complex laws and unnecessary legal formalities simplified?                                |
| Z13  | Legal accountability          | To what extent are rigorous control and supervision exercised by overarching legal authorities and the government?                  |
| Z14  | Professional accountability   | To what extent are citizens adequately guided by professionals to resolve their administrative issues?                              |
| Z15  | Professional accountability   | To what extent are mandated departmental services provided appropriately and to standard?                                           |
| Z16  | Professional accountability   | To what extent is the current staffing level adequate relative to the operational needs of the department?                          |
| Z17  | Professional accountability   | To what extent is clear information regarding performance, policy execution, and departmental programs provided to citizens?        |
| Z18  | Professional accountability   | To what extent do you perceive the administrative services provided to be fair and equitable?                                       |
| Z19  | Professional accountability   | To what extent has this department succeeded in facilitating services and reducing the processing time of administrative workflows? |
| Z20  | Professional accountability   | To what extent do citizens hold trust in the commitments made by the departmental management?                                       |
| Z21  | Political accountability      | To what extent are citizens empowered to participate in or influence the decision-making processes of this department?              |
| Z22  | Political accountability      | To what extent is public or stakeholder feedback integrated into the selection and evaluation of departmental managers?             |

| Item | Construct                | Question                                                                                                                                 |
|------|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| Z23  | Political accountability | To what extent has this department succeeded in eliminating administrative discrimination and ensuring equal societal treatment?         |
| Z24  | Political accountability | To what extent is there a clear alignment between the department's executive programs and the broader political expectations of society? |

Note. Control activities and financial management items were adapted from Wakiriba et al. (2014). Accountability items were adapted from Mulgan (2000), Bovens (2007) and Romzek and Dubnick (1987).